

Forecasting sovereign default risk with Merton's model

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Abstract

Merton's structural model for sovereigns is proven to be useful to analyze the default risk of a country. We are the first to investigate how fast CDS spreads react to changes in model inputs and outputs. CDS spread changes strongly correlate with exchange rate returns, which are an input to the model. But CDS spread changes on average react with a delay to changes in model outputs such as the distance to default, the default probability and model spreads. Hence contingency claim analysis for sovereigns provides useful predictions for CDS spreads.

Keywords: Sovereign credit risk; structural model; emerging debt.

JEL codes: G13, G14, G17

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1. Introduction

In Merton's (1974) structural model the equity of a firm is modeled as a call option on the market value of the assets with a strike price equal to the debt of the firm. As such the model provides a theoretical relationship between equity and corporate bond prices. Gray et al. (2007) adapt Merton's model to apply it at the aggregate level to the sovereign balance sheet. The basic idea is that the local currency debt can be seen as the equity of a sovereign given the ability of a country to create new money. The foreign currency debt is similar to a corporation's debt because a country cannot easily create more foreign currency and practice shows that foreign currency debt is senior to local currency liabilities (Sims, 1999).

Several studies provide an empirical analysis of the sovereign structural model. Gray and Jones (2006), Gray et al. (2007), Keller et al. (2007) and Gapen et al. (2008) test the model on a combined 13 emerging countries. In general these studies find that the model outcomes make sense in that there is a large correlation between model-implied spreads (or the distance to defaults) and market spreads. Kalteier and Posch (2013) use market Credit Default Swap (CDS) data of 14 developed markets to calibrate the model parameters. They find that spreads of higher leveraged countries react more to global equity markets, whereas spreads of less leveraged countries react more to political stability.

In this study we focus on dynamics in spreads that are implied by the sovereign structural model. How long does it take before model implied spread changes and changes in other model outcomes are incorporated into the CDS market? This is different from what existing studies do. For example Gapen et al. (2008) find a –83 percent correlation between distance-to-default and CDS spreads for 12 emerging countries. We, however, only find a correlation of –9 percent between 1-month *changes* in distance to defaults and 1-month *changes* in CDS spreads for 14 emerging countries. The correlation rises to –20 percent when comparing 3-month changes instead of 1-month changes. Hence this is a clear indication that the CDS market is only slowly adapting to changes in the creditworthiness of a country picked up by the sovereign structural model. This is further underscored by our finding that past changes in the distance to default can predict future changes in CDS spreads. The predictive power is significant both statistically and economically.

One possible explanation for the found inefficiency in CDS markets is that the structural model outcomes are invisible to market participants. So what about the

inputs? In the sovereign structural model we need to express the local currency liabilities of a sovereign in US Dollars. Hence there is an important role in the model for both exchange rates and exchange rate volatility. Of course exchange rate returns are directly visible in the market so that should not cause predictive ability. Indeed we do find a strong contemporaneous correlation of about 50 percent between exchange rate returns and changes in CDS spreads, regardless of using the 1-month or 3-month sampling frequency. And indeed we do not find any predictability from exchange rate returns to changes in CDS spreads.

However exchange rate volatility and the interaction with assets and debt in the sovereign structural model are not visible to market participants. In fact we find that model implied spread changes, changes in the probability of default and distance to default can all predict changes in CDS spreads. Even changes in exchange rate volatilities predict changes in CDS spreads. These results echo those of Bharath and Shumway (2008) for corporations. They show that the Merton distance to default measure can predict the actual default probability¹.

There is however also an important difference with the findings for corporations. Gebhardt et al. (2005) find that equity returns predict corporate bond returns of the same firm. They attribute this finding to the debt market being less informational efficient than the equity market, among other factors due to stocks being more actively traded than corporate bonds. Hilscher et al. (2014) report that the equity market is also more efficient than the CDS market. The CDS market is more efficient than the corporate bond market according to Blanco, Brennan and Marsh (2005). The equity in the sovereign structural model is equivalent to the local currency debt expressed in dollars. Hence 'equity returns' depend on local currency

¹ For corporations Avramov et al. (2007) and Zhang et al. (2009) find a significant contemporaneous correlation between changes in equity volatility and changes in CDS spreads. But these studies do not investigate whether changes in equity volatility predict changes in CDS spreads. We also find a significant contemporaneous correlation between changes in exchange rate volatility and changes in CDS spreads. Carr and Wu (2007) find that CDS spreads of Mexico and Brazil co-vary with their respective currency option implied volatilities. They argue that the credit quality of a country is linked to the volatility of the currency because they are both positively linked to a country specific risk factor. The structural model provides such a link through the market value of the country's assets: A rise in exchange rate volatility will reduce the distance to default and hence increase the credit risk.

debt returns and exchange rate returns. These returns, however, cannot predict emerging sovereign debt issued in dollars or changes in CDS spreads.

The remainder of this document is as follows. Section 2 describes the structural model for sovereign countries. Section 3 provides details on the data. The results are presented in Section 4. Finally, Section 5 concludes.

2. Sovereign structural model and testable implications

Gray et al. (2007) adapt Merton's (1974) model for corporations to make it applicable to sovereign countries. They start with formulating the combined balance sheet of the government and the central bank, see Table 1. The assets of the sovereign country consist of four items: (i) Foreign currency reserves including actual reserves and contingent reserves from international financial institutions such as the IMF and other governments; (ii) The net fiscal assets which equal the government's budget surplus or deficit on taxes and revenues minus expenditures; (iii) Credit to other sectors such as the corporate, financial and household sector; and (iv) Other public assets such as the equity of public enterprises.

-Insert Table 1 about here-

The liabilities also contain four items: (i) (Implicit) financial guarantees to the so-called too-big-to-fail entities as we have seen in the government support and bailouts of large financial institutions during the financial crisis of 2008 to 2012; (ii) Foreign currency debt which is issued by the public sector and denominated in a foreign currency; (iii) Local currency debt which is issued by the public sector and denominated in the local currency; and (iv) The monetary base which is related to the money supply in the country's economy. The monetary base consists of the commercial banks' reserves maintained in accounts of the local central bank plus the total currency circulating in the public.

Gray et al. (2007) argue that the balance sheet of the country's public sector and the balance sheet of a corporation show important similarities in both structure and priority of the claims. Local currency debt and the monetary base are called local currency liabilities and have certain features similar to the equity of a corporation. The public sector controls the money supply and therefore they have the option to repay their local currency debt by creating more domestic currency. However,

expansion of the money supply can cause inflation which lowers the real value of the payments to the local currency debt holders. This is similar to equity of corporations because excessive issuance of shares dilutes existing holders' claims and reduces the price per share on the balance sheet of a corporation.

The foreign currency debt is analogous to the risky debt of a corporation because here the public sector cannot easily create more foreign currency since excessively creating domestic currency will lower the demand of the domestic currency and hence depreciate the foreign exchange rates. Moreover, Sims (1999) considers foreign currency debt to be senior to local currency liabilities because in stress situations most governments prefer to inflate local currency debt instead of defaulting on foreign currency debt. Finally, financial guarantees are hard to measure and are therefore not explicitly included in the analysis. Implicitly the guarantees are subtracted from the asset side of the balance sheet, similar to Bodie and Brière (2014).

The sovereign Contingent Claims Approach (CCA) model is similar to the corporate CCA model with local currency liabilities modeled as an implicit call option on the country's assets and foreign currency debt modeled as a distress barrier minus an implicit put option on the country's assets. The distress barrier is based on the book value of the foreign currency debt using the empirical rule that it is equal to the short-term debt plus fifty percent of the long-term debt.

An important feature of the sovereign CCA model is that the balance sheet items are measured in one currency unit, commonly a "hard" currency such as the US Dollar or the Euro, with the corresponding risk-free interest rate. This means that the local currency debt and the monetary base are converted to the US dollar using the market price of the foreign exchange rate.

Essentially, the sovereign CCA model is the same as the corporate CCA model. We get two equations with two unknowns, the market value of the assets (A) and the volatility of the market value of the assets (σ_A). The market value of the equity of the corporation is replaced by the market value of the local currency liabilities (LCL), the equity volatility by the volatility in LCL (σ_{LCL}), and the distress barrier (B_f) is based on the book value of the foreign currency debt. We then get the following two equations:

$$LCL = AN(d_1) - B_f e^{-rT} N(d_2) \quad (1)$$

$$\sigma_{LCL} = \frac{A}{LCL} \sigma_A N(d_1) \quad (2)$$

where

$$d_1 = \frac{\ln(A/B_f) + (r + \frac{1}{2}\sigma_A^2)T}{\sigma_A\sqrt{T}}, \quad d_2 = d_1 - \sigma_A\sqrt{T},$$

r is the risk-free rate of interest and $N(.)$ is the cumulative normal distribution function. To test the model we analyze the contemporaneous correlation between the model implied credit spread and 5-year CDS spread changes. We also use the model spread changes to predict CDS returns in a trading strategy. Ang and Longstaff (2013) argue that the CDS spread is a more direct measure of sovereign credit risk than a spread based on government bonds. Sovereign debt spreads contain more noise from other factors like changes in the supply of the underlying bond and illiquidity effects in sovereign debt prices. To maintain consistency for the maturity of the CDS contracts we set T to be equal to 5 in the model.

The foreign exchange rate is indirectly an important input parameter because it largely influences the value and volatility of the assets and the local currency liabilities. All input parameters are directly observable except for the market value and volatility of the assets of the sovereign which, therefore, can be estimated from equations (1) and (2).

There are a few important risk measures that can be derived from the model output. First, the risk-neutral probability of default of the sovereign is equal to $N(-d_2)$. Second, the distance to default is d_2 . Finally, to find the model-implied credit spread we first need to find the current value of the risky debt with promised payments B_f . The value of the risky debt at time T is exactly the same as B_f minus the payoff of a put option on the assets, with exercise price equal to B_f . Since the liabilities of the sovereign are equal to the assets of the sovereign, we could also derive the value of the risky debt D_f by subtracting the value of the local currency liabilities from the assets:

$$D_f = A - LCL$$

Then the yield-to-maturity of the risky debt is

$$y = \frac{\ln(B_f / D_f)}{T}$$

and the model-implied credit spread is equal to the difference between this yield and the risk-free rate, r ,

$$s = y - r \tag{3}$$

In this study we are interested how fast information from the sovereign CCA model is incorporated in CDS spreads. We formulate two testable implications of the sovereign structural model:

Hypothesis 1: model spreads are correlated with market spreads

The changes of the sovereign credit spread based on the structural model are empirically correlated to the changes of sovereign market spreads during the same time period.

Hypothesis 2: Estimated model spreads contain useful information about future market spread dynamics

The model spread may contain information that is not priced in yet by the sovereign market spread. Using the model spread dynamics we may be able to predict future changes of the market spreads.

In the analyses we will not only focus on changes model spreads but also on changes in the other model outputs, the distance-to-default and the probability of default. We will also compare changes in credit spreads with changes in exchange rate returns. The latter is an input in the sovereign CCA model.

3. Data

We apply the model of Gray et al. (2007) to emerging markets. There are several reasons why the sovereign CCA model is not applicable to developed countries. First, developed countries have direct access to large and liquid international markets to issue debt in their domestic currency and that is why developed countries have no or only a small amount of foreign currency debt. Therefore, the foreign distress barrier

of most developed countries will be zero. Moreover, countries from the Economic and Monetary Union (EMU) have very limited control over the money supply of the European Central Bank (ECB) and therefore the analogy between local currency liabilities and equity disappears. This leads us to the conclusion that emerging countries with a considerable amount of foreign currency debt are the best test cases to apply the sovereign CCA model to.

The empirical analysis includes fourteen emerging market (EM) countries: Argentina, Brazil, Chile, Colombia, Hungary, Indonesia, Malaysia, Mexico, Peru, the Philippines, Poland, South Africa, South Korea and Turkey. This choice is based on the availability of local and external debt as covered by JP Morgan and Barclays indices and the liquidity and nature of the emerging currencies and credit default swap (CDS) contracts: a country enters the dataset when its currency is floating and when the CDS contract is liquid. CDS liquidity is based on the inclusion of countries in the Markit CDX emerging market index. The CDX index is a weighted index of country CDS contracts and rolls semi-annually in March and September. A wide dealer and industry support allows for significant liquidity in all market conditions and the CDX index is accepted as a key benchmark of the overall market credit risk in emerging markets. We use the same dates as De Zwart et al. (2009) to determine when a currency becomes floating. Since 2009 no new relevant emerging currencies became floating².

The seven countries China, Czech Republic, Egypt, India, Slovakia and Thailand do have outstanding local and external debt, but have not been included in the historical composition of the CDX index. The five countries Bulgaria, Venezuela, Panama, and Ukraine which are or have been included in the CDX index are not included in our research because of not having local or external debt listed in the JP Morgan and Barclays indices. Romania was recently added to the local debt index, but is no longer included in the CDX index. Table 2 shows the data availability for the 14 countries included in the analysis.

-Insert Table 2 about here-

² Perhaps an exception could have been made for Russia, which according to the Central Bank of the Russian Federation (BIS Papers No 57, <http://www.bis.org/publ/bppdf/bispap57u.pdf>) has gradually moved towards a floating currency. Including Russia from March 2009 onwards does not materially affect our results and conclusions.

We obtain daily data from Bloomberg for the period from December 2001 to June 2014 on exchange rates, 5-year CDS spreads and US 5-year government bond yields for the risk free discount factor in the model, assuming a 5-year horizon to match the CDS maturity. We collect data on the market value of local currency bonds from the JP Morgan GBI EM broad index (broadest local currency index with longest data history) and the Barclays Emerging Markets Inflation Linked Bond Index for respectively the nominal and inflation-linked bonds. The inclusion of inflation linked bonds is important. According to Swinkels (2012) in 2003 less than 1% of the emerging market local currency bonds were inflation linked, but that percentage increased to more than 14% in 2011. Local currency bond data for South Korea are obtained from the JP Morgan GBI broad index for developed markets³. The local currency liabilities, *LCL*, are the sum of the market values of the local currency nominal and inflation-linked debt. For all countries we express these local liabilities in US dollars. In addition we add the size of the monetary base (M0) based on IMF data⁴. The sum of the *LCL* and M0 act in the same way as the market value of the equity does in the application of the Merton model for corporations.

For the default barrier, B_f , we use the market value of the foreign currency debt. We aggregate the debt in USD and EUR as included in the JP Morgan EMBI global and the Euro EMBIG indices⁵. B_f is also expressed in US Dollar terms.

The final input parameter is the volatility of *LCL*. The volatility is defined as the standard deviation of daily returns of *LCL* over the past three months giving it both some stability but also being able to timely react to new information. Note that this volatility will depend on the volatility of the market value of the local debt and the exchange rate volatility. As one would expect the main driver of the volatility is the exchange rate volatility. One discomfoting aspect of the volatility of *LCL* is that

³ According to the World Bank classification South Korea is considered to be a developed market. The major index providers JP Morgan and FTSE therefore decided to include the country in their developed markets index universe for respectively bonds and equities. Due to accessibility issues and in particular a lack of full currency convertibility two other major index providers Barclays and MSCI consider South Korea as emerging market for respectively bonds and equities.

⁴ For Argentina and South Korea we use data from the local central bank, and for Peru and the Philippines we use M1 instead of M0 since we could not obtain M0 data from Bloomberg. In addition to the monetary base M1 also includes demand deposits, but not savings accounts.

⁵ Book values of the Euro EMBIG country indices are not available, and we use the market value instead. We checked the robustness of results using the book value of the EMBI global mixed with the market value of the Euro EMBIG, results are robust and somewhat stronger. We use the market value of the EMBI global and the Euro EMBIG indices in the remainder of the paper.

there are jumps due to the redemption or issuance of new local debt. For computing the volatility of *LCL* we try to remove such jumps when they are likely connected to redemptions or new issuance by excluding daily changes of *LCL* (in local currency) larger than 5% removing about 0.6% of all observations. We will analyze the sensitivity of our results to the chosen threshold level of 5%.

Table 3 shows the average and standard deviation of the CDS spreads, the amount of *LCL*, B_f and the standard deviation of the exchange rates (FX volatility). The average CDS spread of the fourteen countries is 270 basis points, with the Latin American region having the largest spreads and the Asian region the smallest. Latin America is also the region with on average the relatively largest size of hard currency debt at USD 19.5 billion and the lowest ratio of local divided by hard currency debt of 4.8. The EMEA region has the largest average currency volatility at 12.0%, compared to 8.8% for Latin-America and 8.6% for Asia. EMEA also has the largest monetary base of 47.1 billion USD on average.

-Insert Table 3 about here-

4. Testing the implications of the sovereign structural model

In Section 4.1 we report on the contemporaneous correlations between CDS spread changes and changes in variables based on the structural model. The outcomes of various trading strategies based on the structural model are presented in Section 4.2. We prefer using trading strategies over predictive regressions following the arguments of Thornton and Valente (2012)⁶. Moreover applying a standard regression approach is not straightforward for the changing composition of our dataset with countries entering and leaving over time.

4.1 Correlations between CDS spread changes and structural variables

We first analyze the correlations between 1-month and 3-month changes in model inputs and outputs with 1-month and 3-month changes in market CDS spreads. By looking at two different frequencies we can observe whether input for and information from the sovereign CCA model is immediately incorporated in market CDS spreads

⁶ Thornton and Valente show that the Cochrane and Piazzesi's (2005) forward rates factor, found to be successful in in-sample predictive regressions for excess bond returns, actually has no economic value in a true real-time exercise.

or with a delay. A similar correlation at the two frequencies suggests an immediate response of market CDS spreads, whereas a higher correlation at the 3-month frequency suggests market CDS spreads only slowly adopt information from the CCA model. The results are shown in Table 4. As expected, the changes in the distance-to-default, d_2 from equation (1), are negatively correlated with the changes in CDS spreads. A larger distance-to-default implies a lower probability of default and hence a lower CDS spread. As expected the changes in the model-implied probability of default, $N(-d_2)$, are positively correlated with changes in CDS spreads. The positive correlations between changes in exchange rate volatility and *LCL* volatility (of the *LCL* expressed in local currency) with CDS spreads are also logical as an increase in either the currency or the interest rate volatility makes it more likely the distance-to-default can be bridged and hence increases the probability of default, all else equal. The negative correlation between changes of the local debt and CDS spreads also makes sense given the lower implied probability of default for a higher level of the local debt. Finally, the positive correlation between CDS spreads and FX returns (measured as local currency units expressed in US dollars) is as expected given that a weaker currency is linked to a higher CDS spread.

-Insert Table 4 about here-

Looking at Panel A and B in Table 4 it is clear that correlations are rising when increasing the window from 1 to 3 months implying noise in the shorter windows and possibly a delayed response of either model-variables or market CDS spreads. We will investigate the latter in the next section. We get the highest positive correlations at 51 and 33 percent between and 3-month changes in CDS spreads and respectively the 3-month spot return of the exchange rates and the changes in exchange rate volatility. Hence a substantial amount of the change in CDS spreads can be explained by changes in the level and volatility of exchanges rates. The level of the correlation between CDS spread changes and FX volatility changes is similar to the 36 percent correlation found in Avramov et al. (2007) between changes in corporate bond spreads and changes in idiosyncratic equity volatility. The 3-month correlation between changes of the distance to default and the CDS spread is also -20% and significant. On a one month horizon the correlation is insignificant though,

indicating a potential delayed response of either the distance to default or the market CDS spread. The correlations with the model spread, the model implied probability of default and the change in the local current debt volatility are not significantly different from zero for both the 1-month and 3-month changes for the portfolio of all countries. We conclude that the first hypothesis that model spreads are correlated with market spreads must be rejected. Perhaps the information in the model spread is able to forecast the market spread. We will investigate that question in the next section.

4.2 Trading strategies

Analogous to the analysis of Gebhardt et al. (2005) for corporations we test whether the Merton model can *forecast* CDS spread changes for countries. The test comprises two different trading strategies.

4.2.1 Relative strategies

In the trading strategies we will predict the relative movement of the CDS spreads for each pair of countries. Reason for a relative strategy is that we want to benefit from the dynamics predicted by the Gray et al. model while hedging against a market-wide spread widening or tightening as a consequence of global (event) risk (see Pan and Singleton, 2008) or global risk appetite or changing global economic conditions (see Baek et al., 2005; Weigel et al., 2006; and Longstaff et al., 2011). The correlation between monthly changes of CDS spreads between countries is 59% on average and positive for all regions⁷.

We use 3-month changes of the seven model in- and outputs analyzed in Table 4 to predict the relative return of two CDS contracts in the next month. For example, if the exchange rate volatility in Brazil has declined more than in Hungary, we expect the CDS spread of Brazil to decline relative to the CDS spread of Hungary. We therefore take a short position in the Brazilian CDS (selling default protection), and a long position in the Hungarian CDS (buying default protection). Since we have data for fourteen countries, each country is compared to thirteen other countries leading to a potential amount of 91 unique relative trades in total.

⁷ This is similar to the finding of a large common component for corporate bonds by Collin-Dufresne and Goldstein (2001) and Elton et al. (2001).

Ben Dor et al. (2007a; 2007b) show that the risk of a CDS contract depends on the level of the spread times the duration. Given that we enter 5-year CDS contracts for both countries and hence durations are approximately equal we simply scale the positions by the CDS spreads. If, for example, the CDS spread is 300 basis points for Brazil and 50 basis points for Hungary this scaling implies the position in the CDS for Hungary is 6 times as large as the one for Brazil. Added benefit is that the periodic spread payments add up to zero as we have a matching long and short position. Hence the profit of the trade only depends on the spread changes in both countries and we create a zero-investment strategy. In equation form we get the following expression for the 1-month return in month t from the opposing CDS positions in countries i and j ,

$$Return_t^{ij} \approx D_{t-1} position_{t-1} \left[\left(\frac{CDS_t^i - CDS_{t-1}^i}{CDS_{t-1}^i} \right) - \left(\frac{CDS_t^j - CDS_{t-1}^j}{CDS_{t-1}^j} \right) \right] \quad (4)$$

where D_{t-1} is the duration (5 years), $position_{t-1}$ is a scaling factor related to the total position size, and CDS_t^i is the CDS spread in month t in country i . For the Sharpe ratio the total position size does not matter. The risk-scaling to match the risks (and payments) of the two CDS positions is reflected in the denominators in equation (4).

For the position size we can take simple long/short positions ($position_{t-1} = \pm 1$), but alternatively we can let the position size depend on the magnitude of the changes in the predictive variables over time. The larger the change, the more spread movement we can expect. We set the position size proportional to a standardized value by computing the current difference in 3-month changes of for example model spreads or exchange rate volatilities, deducting its own average over the previous 5 years and dividing by its own 5-year standard deviation.

4.2.2 Relative strategy results

Table 5 shows the Sharpe ratios for the CDS strategy based on 3-month changes of the seven model in- and outputs analyzed in Table 4. The results are positive for all seven indicators. The best indicator is the change of the model spread with a Sharpe ratio of 0.52. Results are strong for EMEA, but weaker for Latin America and Asia. An explanation may be the lower currency volatility. Also, results for one region may

be influenced by the other regions because all positions in one country are vis-à-vis another country. In the next section we investigate the results for individual countries.

This strong result proves that the sovereign CCA model of Gray et al. is a nice step towards better understanding sovereign credit risk. Furthermore this result shows that the market does not fully incorporate information from the aggregate sovereign balance sheets in the CDS market. The low contemporaneous correlation between the changes in model spreads and market spreads in Table 4 already indicated that the market does not react directly on the information embedded in the model spread.

The strategy on basis of changes in exchange rate volatility (σ_{FX}) has a Sharpe ratio of 0.48 underscoring the importance of currency volatility for CDS spreads. Currency volatility was already important in the contemporaneous correlation analysis in Table 4, but also important here in predicting changes in market CDS spreads.

Finally, the FX factor (the spot return of local currency units expressed in US dollars) achieves the weakest Sharpe ratio of 0.11. The FX factor also shows the highest correlation with the CDS market spread changes in Table 4. The high correlation may indicate that direct FX returns are priced in more efficiently by the CDS market, contrary to the information embedded in the model spread. The results support the second hypothesis that indicators based on the model contain useful information about future sovereign default spread changes.

-Insert Table 5 about here-

4.2.3 Directional strategies

The directional strategy intends to profit from the movement of the CDS spread of an individual country. Contrary to the relative strategy, the directional strategy is subject to a market-wide spread widening or tightening and the return of the directional strategy depends on the carry of the CDS contract.

For the direction of the trade we look at 3-month changes in the same seven model in- and outputs. For example, if the exchange rate volatility in Brazil has declined we expect the Brazilian CDS spread to decline as well. We therefore take a short position in the Brazilian CDS contract (selling default protection).

Similar to the relative strategy the position size is proportional to the CDS spread in order to obtain a similar risk per country. In equation form we then get the

following expression for the 1-month return in month t from the CDS position (long position is buying default protection) in country i ,

$$Return_t^i \approx \frac{position_{t-1}}{CDS_{t-1}^i} \left[D_{t-1} (CDS_t^i - CDS_{t-1}^i) - \frac{CDS_{t-1}^i}{12} \right] \quad (5)$$

where D is the duration (5 years), $position_{t-1}$ is a scaling factor related to the total position size, and CDS_t^i is the CDS spread in month t in country i . For the success ratio and Sharpe ratio the total position size does not matter. Because the size of the position is adjusted for the level of the CDS spread, the carry return is the same for each month and each country: the position size divided by 12.

4.2.4 Directional strategy results

Table 6 shows the Sharpe ratios for the CDS strategy based on changes in the seven model in- and outputs. Except for the directly observable sovereign CCA model inputs, local debt and exchange rate returns, the results are strong for all other model in- and outputs with Sharpe ratios between 0.43 and 0.54. Compared to the relative strategy the Sharpe ratio of the local interest rate volatility has a much higher Sharpe ratio of 0.54. Contrary to the relative strategy Sharpe ratios are more positive for Latam and Asia. The strategy does not suffer from the larger risk exposure to market-wide changes and is more stable over the regions.

Again the explanation for the weaker predictive results for local debt and currency returns is the higher contemporaneous correlation we find in Table 4. Hence directly observable quantities are efficiently incorporated in market CDS spreads. But the not directly observable currency volatility and all sovereign CCA outputs are not efficiently incorporated in CDS spreads.

Hence also the directional strategies support the second hypothesis that the change in the model spread and its related indicators contain important information for future spread changes.

-Insert Table 6 about here-

4.3 Robustness analyses

We have investigated the sensitivity of the results for the threshold level we use to omit redemptions or issuance of new local debt. In addition we analyze whether the investment strategy returns are significant and not subsumed by the term, default and liquidity premia in bond markets.

4.3.1 Sensitivity analysis threshold level

We have analyzed the sensitivity of the results⁸ to the threshold level we use to omit large daily changes in the *LCL* input factor. The default threshold level has been chosen at a level of 5% removing about 0.6% of all observations. We analyze the results of different threshold levels for both the relative and directional trading strategies based on the three month change of the model spread.

Large changes of the *LCL* may have been caused by a redemption or issuance of new local debt and are not necessarily a sign of higher risk. Therefore, removing these jumps may improve the effectiveness of the model in predicting future CDS spread changes. Since we cannot directly observe redemptions and new issuance in our database we simply omit extreme absolute changes of the *LCL*.

Results are shown in Figure 1. The highest Sharpe ratios are obtained at the threshold level of 8% for both the relative and directional strategies. Without a threshold (far right in Figure 1) the directional strategy becomes weaker with a Sharpe ratio of 0.22, while the relative strategy still obtains a reasonable Sharpe ratio of 0.36. With a low threshold (far left in Figure 1) of 1% about 7% of observations are omitted but the predictive power of the model is robust. A low threshold also omits large moves of the interest rate markets, muting the volatility and leaving out valuable information from the model. The default setting of 5% seems reasonable and results are robust for this choice.

-Insert Figure 1 about here-

4.3.2 Regression analysis

Gebhardt et al. (2005) find that the term premium and default premium are priced risk factors for corporate bond returns. Longstaff (2004) finds a priced liquidity measure

⁸ We focus on the results for the trading strategies. The correlations in Table 4 are also robust for the threshold level but are not included here to save space. Results are available upon request.

based on the market cap weighted excess return of RefCorp⁹ compared to US treasuries. We run the following regression of the returns of the relative and directional strategies on the default premium, term premium and liquidity premium:

$$R_t = \alpha + \beta_1 DEF_t + \beta_2 TERM_t + \beta_3 LIQ_t + \varepsilon_t \quad (6)$$

where R_t are the relative or directional strategy returns based on the seven model in- and outputs analyzed in Table 4, α the unexplained return, DEF_t the default premium, $TERM_t$ the term premium and LIQ_t the liquidity premium. Over the sample period from December 2001 to June 2014 the term premium is 2.83% per annum, the default premium 5.98% per annum and the liquidity premium -0.03% per annum. The results of the regression in equation (6) are presented in Tables 7 (relative strategy return) and 8 (directional strategy returns).

-Insert Table 7 and Table 8 about here-

For the relatively strategies in Table 7 only the ones based on currency returns and changes in currencies volatility have a significant negative loading on the default premium. Given the positive default premium this actually adds to the alpha but these alphas are not significantly different from zero. The relative strategy based on the changes in the volatility of local debt returns loads significantly on the term premium which reduces alpha. The only statistically significant alpha (p-value 0.02) is the one for the relatively strategy based on the distance-to-default. The returns of this strategy do not load on the default and term premium, and the alpha is equal to 2.4% per annum. The other model outcomes, spreads and probabilities of default, have similar alphas as the distance-to-default but these are not significant. One possible explanation is that the distance to default can take values with a more uniform distribution compared to the model spread and probability of default. The probability of default is equal to zero in 44% of the observations and has a maximum of 9.3%. The skewness of the probability of default is also extreme with a value of 13.3,

⁹ RefCorp is short for the Resolution Trust Corporation which is a US government corporation established to rescue savings and loan institutions that failed during the savings and loan crisis. RefCorp provided liquidity to these organizations by issuing bonds. RefCorp's bonds have the same standing as Government bonds in case of a credit event. Hence price differences are only due to liquidity differences.

whereas the skewness of the distance to default is more benign with a value of 3.0. The distance to default is therefore better able to distinguish a safe country from a safer country.

For the directional strategies in Table 8 only the one based on past local debt returns significantly loads on the default premium with a coefficient of -0.43. This benefits the alpha but it does not overcome the large negative total return of -4.2% in Table 6. In terms of the alphas all but the ones based on currency and local debt returns are significant at least at the 10% significance level. Also here the directional strategy based on the distance-to-default is the most significant with a p-value of 0.01. Its (insignificant) negative loadings on the default and term premia make sense since the strategy invests in countries with a declining implied credit risk.

Hence corrected for the market and default premiums not all information from the aggregate sovereign balance sheets is fully incorporated in the CDS market. The strongest significance is visible for the strategies based on changes in the distance-to-default.

5. Conclusion

We provide an extensive empirical study into the Gray et al. (2007) structural model for sovereigns. We show that the structural approach for emerging countries that issue both local and foreign currency denominated debt has its merits. In the model the exchange rate dynamics play an important role. Specifically the model predicts a close link between exchange rate returns and volatility changes on the one hand, and market CDS spread movements on the other hand.

The model outcomes such as the distance-to-default, the default probability, and spreads are strongly correlated with market CDS spreads. Also exchange rate dynamics, an important input for these model outcomes, are strongly correlated with market CDS spreads. This is a similar role to what other studies find for equity returns and volatility in Merton's structural model for corporations. The results reject our first hypothesis that the changes of the sovereign credit spreads based on the structural model should be empirically correlated to changes of sovereign market spreads during the same time period.

The second hypothesis is that information embedded in the model spreads may contain information that is not priced in yet by the sovereign market spreads. Because the currency market is more efficient than the sovereign CDS market, exchange rate

information could have predictive power for future CDS spread changes. We build strategies based on the model spreads that indeed show a strong and significant performance. This indicates that the market does not fully price the sovereign balance sheet information into CDS spreads.

For further research it will be interesting to look for improvements to the Merton framework applied to sovereigns. The structural model may be improved with suggestions from the study of Eom et al. (2004) on corporations regarding the use of a different model set-up allowing for a stochastic interest rate and a recovery at default among others. In addition Aktug (2014) who applies the Merton model to three emerging markets argues that more precise data on the debt and the addition of other volatility measures like the equity market volatility may help.

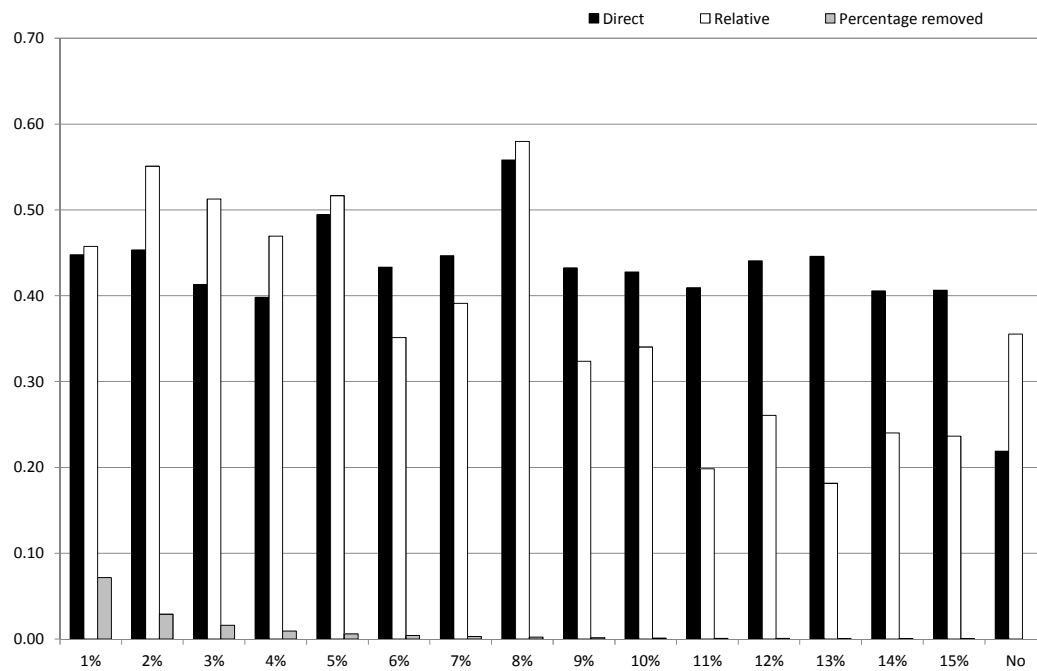
Finally, including alternative factors in a reduced form model set-up like Ang and Longstaff (2013) use for EMU countries is another possibility for further research.

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Figure 1: Results strategies based on the model spread



Note: the LCL input of the model contains jumps due to the redemption or issuance of new local debt. In the calculation of the volatility of the LCL we try to remove such jumps by excluding daily changes of LCL larger than 5%, removing about 1% of all observations. We show the sensitivity of the Sharpe ratio (y-axis) of the relative and direct trading strategies based on three month changes in model spreads for different threshold levels (shown on x-axis) over the period from December 2001 to June 2014. The Sharpe ratios of the trading strategies and the percentage of observations are shown in bars.

Table 1: Combined balance sheet of the government and the central bank

Assets	Liabilities
Foreign reserves	Guarantees
Net fiscal assets	Foreign currency debt
Credit to other sectors	Local currency debt
Other public assets	Monetary base

Note: the equity of the country is represented by the sum of the local currency debt and the monetary base which we call local currency liabilities (LCL). The debt of the country is represented by the foreign currency debt. Financial guarantees are hard to measure and are therefore not explicitly included in the analysis. Implicitly the guarantees are subtracted from the asset side of the balance sheet, similar to Bodie and Brière (2014).

Table 2: Data availability and regions

	First month	Last month	Region
Argentina	Sep 2005	Jun 2014	Latam
Brazil	May 2002	Jun 2014	Latam
Chile	Mar 2014	Jun 2014	Latam
Colombia	Jan 2003	Jun 2014	Latam
Hungary	Mar 2009	Mar 2013	EMEA
Indonesia	Mar 2006	Jun 2014	Asia
Malaysia	Jul 2005	Jun 2014	Asia
Mexico	Jan 2002	Jun 2014	Latam
Peru	Oct 2006	Jun 2014	Latam
the Philippines	Oct 2010	Jun 2014	Asia
Poland	Feb 2004	Sep 2004	EMEA
South Africa	Dec 2001	Jun 2014	EMEA
South Korea	Feb 2002	Apr 2004	Asia
Turkey	Apr 2004	Jun 2014	EMEA

Note: the data availability is based on the availability of local debt, external debt, the monetary base, the composition of the CDX index and whether the country has a fully floating FX rate. Hungary is dropped from the CDX index in April 2013, Poland is dropped from the CDX index in October 2004 and South Korea is dropped from the JP Morgan EMBI global index in May 2004 (and in March 2006 also from the CDX index).

Table 3: Average and standard deviation spreads, debt and FX volatility

		Latam	EMEA	Asia	Portfolio
CDS	Avg	375.5	228.9	153.9	270.3
spread (bp)	St dev	438.0	156.6	96.2	302.3
LCL	Avg	110.7	100.5	91.9	101.5
bln USD	St dev	141.2	65.4	75.5	93.7
B _f	Avg	19.5	10.3	8.2	12.2
bln USD	St dev	17.0	10.4	6.3	12.0
Ratio LCL / B _f	Avg	4.8	17.9	14.4	12.6
bln USD	St dev	2.5	13.8	20.2	14.1
M0	Avg	39.2	47.1	30.3	39.8
bln USD	St dev	53.2	49.0	10.7	41.9
	Avg	8.8%	12.0%	8.6%	9.9%
FX volatility	St dev	3.6%	4.4%	4.1%	4.2%

Note: Sample averages and standard deviations CDS spread (basis points), local currency liabilities in bln USD (LCL), default barrier in bln USD (B_f), the ratio between LCL and B_f, monetary base in bln USD (M0) and the annualized FX volatility from December 2001 - June 2014. The final column is the equal weighted average of the fourteen countries. Latam (Latin America) is the equal weighted portfolio of Argentina, Brazil, Chile, Colombia, Mexico and Peru. EMEA includes Hungary, Poland, South-Africa and Turkey. Asia includes Indonesia, Malaysia, The Philippines and South-Korea.

Table 4: Correlations of CDS spread changes with model inputs and outputs

	Latam	EMEA	Asia	Portfolio
<i>Panel A: 1-month changes</i>				
Spread	-1%	1%	3%	0%
N(-d ₂)	0%	2%	3%	1%
d ₂	-12%	-10%	-5%	-9%
σ _{FX}	21%**	21%**	20%**	20%**
FX	49%***	57%***	49%***	49%***
σ _{Debt}	8%	-2%	5%	5%
Local debt	-22%***	-10%	-4%	-15%*
<i>Panel A: 3-month changes</i>				
Spread	9%	9%	14%*	10%
N(-d ₂)	12%	11%	15%*	12%
d ₂	-18%**	-31%***	-20%**	-20%**
σ _{FX}	30%***	37%***	38%***	33%***
FX	50%***	60%***	46%***	51%***
σ _{Debt}	13%	15%*	13%	13%
Local debt	-30%***	-10%	-2%	-20%**

Note: Correlations between the relative change in 5-year CDS spreads and (i) the change in the model-implied spread from equation (3); (ii) the change in the model-implied probability of default, N(-d₂); (iii) the change in the model-implied distance-to-default, d₂; (iv) the change in the exchange rate volatility, σ_{FX}; (v) FX, the spot return of local currency units expressed in US dollars; (vi) the change in the LCL volatility expressed in the local currency, σ_{Debt}; (vii) and Local debt, the relative change in the value of the local currency liabilities expressed in local currency. In the upper panel we look at 1-month changes, in the lower panel at 3-month changes. Latam is short for Latin America, and represents the equal weighted portfolio of Argentina, Brazil, Chile, Colombia, Mexico and Peru. EMEA includes Hungary, Poland, South-Africa and Turkey. Asia includes Indonesia, Malaysia, The Philippines and South-Korea. The sample period is December 2001 to June 2014 resulting in 151 monthly observations. Overlapping data are used for 3-month changes, every time shifting time by one month. The 1%, 5% and 10% significance levels the correlations are denoted by ***, ** and *. The test is based on the standard significance test for correlation $Test = \frac{r}{\sqrt{(1-r^2)(n-2)}}$, where r is the correlation and n is the number of observations. The significance of the test depends on the t-distribution with $n - 2$ degrees of freedom.

Table 5: CDS relative strategy

	Latam	EMEA	Asia	Portfolio	Return	p-value
Spread	-0.06	0.71	0.19	0.52	3.0%	0.13
$N(-d_2)$	-0.09	0.75	0.09	0.49	2.8%	0.15
d_2	0.25	0.63	-0.07	0.48	2.2%	0.02
σ_{FX}	-0.17	0.76	-0.01	0.25	1.0%	0.42
FX	0.02	0.21	-0.40	-0.04	-0.2%	0.88
σ_{Debt}	0.04	0.65	-0.09	0.29	1.6%	0.28
Local debt	0.15	0.03	0.27	0.28	1.4%	0.30

Note: Sharpe ratios for the relative strategy trading market CDS country spreads based on seven model in- and outputs (see Table 4). For the portfolio the return and the Newey-West corrected p-values have been added. At the end of each month we compute the 3-month change in the model spread for each country. For each country pair we compute the difference in the 3-month changes, as well as the 5-year average and the 5-year standard deviation of these differences. The resulting z-score is used as position size in equation (4). The movements in the CDS spreads in the next month (t) are used to compute the performance of the trade using equation (4). We follow this strategy from December 2001 to June 2014. The table shows the Sharpe ratios of the three regions and the average portfolio of all the fourteen countries. Latam is short for Latin America, and represents the equal weighted portfolio of Argentina, Brazil, Chile, Colombia, Mexico and Peru. EMEA includes Hungary, Poland, South-Africa and Turkey. Asia includes Indonesia, Malaysia, The Philippines and South-Korea.

Table 6: CDS directional strategy

	Latam	EMEA	Asia	Portfolio	Return	p-value
Spread	0.22	0.66	0.41	0.49	10.7%	0.13
$N(-d_2)$	0.21	0.69	0.32	0.48	10.3%	0.13
d_2	0.21	0.67	0.25	0.48	6.3%	0.04
σ_{FX}	0.13	0.57	0.27	0.34	6.3%	0.20
FX	0.05	0.00	-0.39	-0.07	-1.6%	0.77
σ_{Debt}	0.25	0.72	0.21	0.54	6.5%	0.07
Local debt	-0.16	-0.29	-0.31	-0.31	-4.2%	0.39

Note: Sharpe ratios from the directional strategy trading market CDS based on seven model in- and outputs (see Table 4). For the portfolio the return and the Newey-West corrected p-values have been added. At the end of each month we compute the 3-month change in the model spread. We compute the average and the standard deviation of these 3-month changes in the past 5 years. The resulting z-score is used as position size in equation (5). The movement in the CDS spread in the next month (t) is used to compute the performance of the trade using equation (5). We follow this strategy from December 2001 to June 2014. The table shows the Sharpe ratios of the three regions and the average portfolio of all the fourteen countries. Latam is short for Latin America, and represents the equal weighted portfolio of Argentina, Brazil, Chile, Colombia, Mexico and Peru. EMEA includes Hungary, Poland, South-Africa and Turkey. Asia includes Indonesia, Malaysia, The Philippines and South-Korea.

Table 7: Regression relative strategy

	α	$p - \text{value } \alpha$	β_1	β_2	β_3	R^2
Spread	2.4%	0.14	0.02	0.14	-0.19	2%
$N(-d_2)$	2.3%	0.17	0.03	0.14	-0.19	2%
d_2	2.4%	0.02	-0.04	0.03	-0.05	1%
σ_{FX}	1.5%	0.20	-0.12***	0.07	0.16	10%
FX	0.7%	0.51	-0.16***	0.00	-0.25**	10%
σ_{Debt}	1.2%	0.42	0.00	0.14*	0.05	2%
Local debt	1.5%	0.28	-0.02	0.00	-0.26*	1%

Note: This table presents the alphas of the regression in equation (6),

$$R_t = \alpha + \beta_1 DEF_t + \beta_2 TERM_t + \beta_3 LIQ_t + \varepsilon_t$$

R_t is the return of the portfolio formed by the relative strategies (see table 5 for more information and details), DEF is the return of the EMBI+ market index over US Treasuries. $TERM$ is the return of US Treasuries over cash. LIQ is the excess return of RefCorp bonds over maturity matched US Treasuries. The constant (α) is expressed per annum, its p-value is denoted in the second column. The 1%, 5% and 10% significance levels for β_1 and β_2 are denoted by ***, ** and *, respectively. We use Newey-West standard errors to correct for heteroscedasticity and autocorrelation. The sample period is from December 2001 to June 2014.

Table 8: Regression directional strategy

	α	$p - \text{value } \alpha$	β_1	β_2	β_3	R^2
Spread	12.4%	0.05	-0.07	-0.46	-1.37	2%
$N(-d_2)$	12.0%	0.05	-0.07	-0.44	-1.30	2%
d_2	7.9%	0.01	-0.19	-0.21	0.45	2%
σ_{FX}	9.5%	0.04	-0.33	-0.51*	0.19	4%
FX	6.2%	0.31	-0.96***	-0.84**	0.80	20%
σ_{Debt}	6.0%	0.10	0.08	0.00	0.04	0%
Local debt	-1.8%	0.65	-0.43**	0.01	0.41	10%

Note: This table presents the alphas of the regression in equation (6),

$$R_t = \alpha + \beta_1 DEF_t + \beta_2 TERM_t + \beta_3 LIQ_t + \varepsilon_t$$

R_t is the return of the portfolio formed by the directional relative strategy (see table 6 for more information and details), DEF is the return of the EMBI+ market index over US Treasuries. $TERM$ is the return of US Treasuries over cash. LIQ is the excess return of RefCorp bonds over maturity matched US Treasuries. The constant (α) is expressed per annum, its p-value is denoted in the second column. The 1%, 5% and 10% significance levels for β_1 and β_2 are denoted by ***, ** and *, respectively. We use Newey-West standard errors to correct for heteroscedasticity and autocorrelation. The sample period is from December 2001 to June 2014.